Blessed G/Michael Catholic Secondary School

Semester 1 Biology Final Examination for Grade12

Academic Year: 2024 / 2018 E.C. Time Allowed: 2:00'

Name: _____ **Grade & Sec:** Grade12 _____ **R.No:** _____

I. Write 'True' if the statement is correct and 'False' if it is incorrect.

____ **1.** Glycolysis occurs in the cytosol of the cell and requires molecular oxygen (O2) to proceed.

____ **2.** In DNA replication, the leading strand is synthesized continuously in the 5' to 3' direction.

____ **3.** Restriction endonucleases are bacterial enzymes that cut DNA molecules at specific palindromic recognition sequences.

____ **4.** Photolysis of water occurs during the light-independent Calvin cycle reactions inside the stroma of chloroplasts.

____ **5.** Fixed action patterns (FAPs) are innate, stereotypic behavioral sequences triggered by a sign stimulus or releaser.

II. Choose the best answer among the given alternatives.

____ **6.** Which of the following is the final electron acceptor in mitochondrial aerobic respiration?

A. NAD+ **B.** Cytochrome c

C. Molecular oxygen (O2) **D.** Pyruvate

____ **7.** During the Krebs cycle (citric acid cycle), how many molecules of NADH are produced per single turn from one acetyl-CoA molecule?

A. 1 NADH **B.** 2 NADH **C.** 3 NADH **D.** 4 NADH

____ **8.** In C4 plants such as maize and sorghum, the initial fixation of atmospheric carbon dioxide is catalyzed by:

A. RuBisCO in mesophyll cells **B.** PEP carboxylase in mesophyll cells

C. RuBisCO in bundle sheath cells **D.** ATP synthase in thylakoid membranes

____ **9.** Which enzyme is responsible for sealing nicks and joining Okazaki fragments in the lagging strand of replicating DNA?

A. DNA Helicase **B.** DNA Ligase **C.** RNA Primase **D.** Topoisomerase

____ **10.** During protein biosynthesis, peptide bond formation between incoming amino acids in the ribosome is catalyzed by:

A. Aminoacyl-tRNA synthetase **B.** Peptidyl transferase (ribozyme)

C. RNA polymerase II **D.** Elongation factor EF-Tu

____ **11.** The anticodon on a tRNA molecule that corresponds to the start codon AUG is:

A. 3'-UAC-5' **B.** 5'-UAC-3' **C.** 3'-TAC-5' **D.** 5'-AUG-3'

____ **12.** Which type of mutation involves the insertion or deletion of nucleotides that shifts the genetic reading frame?

A. Silent mutation **B.** Missense mutation

C. Frameshift mutation **D.** Nonsense mutation

____ **13.** In the polymerase chain reaction (PCR), what is the primary purpose of heating the reaction mixture to approximately 94°C–96°C?

A. Annealing primers to template DNA **B.** Elongation of target DNA strands

C. Denaturation of double-stranded DNA **D.** Activating the Taq DNA polymerase

____ **14.** Which vectors are most commonly utilized to transfer foreign recombinant genes into plant genomes, such as in creating GM crops?

A. Bacteriophage lambda
B. Ti plasmid of Agrobacterium tumefaciens
C. SV40 mammalian virus
D. pBR322 human artificial chromosome

____ **15.** A cross between a homozygous dominant purple flower (PP) and a homozygous recessive white flower (pp) produces F1 heterozygous purple flowers (Pp). If F1 plants self-pollinate, what is the phenotypic ratio in the F2 generation?

A. 1 purple : 1 white
B. 3 purple : 1 white
C. 9 purple : 3 white : 3 red : 1 pink
D. 1 purple : 2 intermediate : 1 white

____ **16.** According to the operon model in Escherichia coli, what binds to the operator region to prevent transcription of structural genes in the lac operon?

A. CAP-cAMP complex
B. Active repressor protein
C. Allolactose inducer
D. RNA polymerase holoenzyme

____ **17.** Which of the following structures is homologous to the wing of a bird?

A. The wing of an insect
B. The flipper of a whale
C. The tail fin of a shark
D. The tentacle of an octopus

____ **18.** In population genetics, which condition is REQUIRED to maintain Hardy-Weinberg genetic equilibrium in a population?

A. High rate of gene flow
B. Frequent beneficial mutations
C. Small population size
D. Random mating within the population

____ **19.** Speciation that occurs when populations are geographically isolated by physical barriers such as mountain ranges or rivers is called:

A. Sympatric speciation
B. Allopatric speciation
C. Parapatric speciation
D. Peripatric speciation

____ **20.** A decreasing response to a repeated stimulus that carries neither reward nor punishment is an example of:

A. Classical conditioning
B. Operant conditioning
C. Habituation
D. Imprinting

____ **21.** Konrad Lorenz showed that newly hatched goslings follow the first moving object they see during a critical period. This form of learning is known as:

A. Imprinting
B. Latent learning
C. Trial-and-error learning
D. Insight learning

____ **22.** In classical conditioning, Pavlov demonstrated that dogs salivate to the ringing of a bell after pairing the sound with food. Here, the bell sound acts as the:

A. Unconditioned stimulus
B. Conditioned stimulus
C. Unconditioned response
D. Operant reinforcer

____ **23.** Which of the following hormones stimulates ripening in climacteric fruits such as bananas and tomatoes?

A. Auxin **B.** Gibberellin **C.** Ethylene **D.** Abscisic acid

____ **24.** During anaphase I of meiosis, what homologous structures separate and migrate toward opposite poles?

A. Sister chromatids
B. Homologous chromosome pairs (bivalents)
C. Non-homologous chromosomes
D. Centrioles

____ **25.** The chemical energy released during oxidative phosphorylation is directly coupled to ATP synthesis by:

A. The flow of protons down their electrochemical gradient through ATP synthase
B. Direct substrate-level phosphorylation of glucose
C. The active pumping of sodium ions into the mitochondrial matrix
D. The breakdown of carbon dioxide in the matrix

____ **26.** In gel electrophoresis, DNA fragments separate based primarily on differences in:

A. Electric charge alone
B. Molecular weight and fragment length
C. Base composition (ratio of A-T to G-C)
D. Isoelectric point

____ **27.** A man with blood type A (heterozygous IAi) marries a woman with blood type B (heterozygous IBi). What blood types are possible among their children?

A. Only type AB
B. Only types A and B
C. Types A, B, and AB only
D. Types A, B, AB, and O

____ **28.** Which of the following is an example of an organelle that contains its own circular DNA and 70S ribosomes?

A. Golgi apparatus
B. Lysosome
C. Mitochondria
D. Endoplasmic reticulum

____ **29.** In eukaryotic pre-mRNA processing, which modifications are required before export to the cytoplasm for translation?

A. Addition of a 5' methyl-guanosine cap, 3' poly-A tail, and splicing of introns
B. Removal of exons and retention of non-coding introns
C. Binding of reverse transcriptase and cleavage of telomeres
D. Phosphorylation of histone octamers

____ **30.** Which of the following evolutionary forces is primarily responsible for introducing novel alleles into a gene pool?

A. Genetic drift
B. Natural selection
C. Mutation
D. Assortative mating

____ **31.** Altruistic behavior among social insects (such as sterile worker honeybees defending the queen) is best explained by:

A. Group selection alone
B. Kin selection and inclusive fitness
C. Operant conditioning
D. Directional selection for body size

____ **32.** The light-dependent reactions of photosynthesis yield two vital energy-rich products used directly in the Calvin cycle:

A. Glucose and O2
B. ATP and NADPH
C. FADH2 and Pyruvate
D. NADP+ and ADP

____ **33.** Sickle-cell trait provides resistance against severe malaria in heterozygous individuals (HbA HbS). This survival advantage exemplifies:

A. Heterozygote advantage (balancing selection)
B. Directional selection
C. Disruptive selection
D. Sexual selection

____ **34.** Which type of learning involves solving a novel problem through sudden understanding of relationships without prior trial-and-error?

A. Habituation
B. Insight learning
C. Classical conditioning
D. Sensitization

____ **35.** In the lac operon of E. coli, when both glucose and lactose are abundant in the growth medium:

A. The lac operon is transcribed at its maximum possible rate
B. cAMP levels are low, preventing high-level transcription (catabolite repression)
C. The repressor binds irreversibly to the promoter
D. Beta-galactosidase synthesizes glucose from starch

____ **36.** Which of the following cellular processes generates the highest net yield of ATP molecules per oxidized mole of glucose?

A. Alcoholic fermentation
B. Lactic acid fermentation
C. Glycolysis
D. Mitochondrial oxidative phosphorylation

____ **37.** Recombinant human insulin is commercially produced in bulk using genetically engineered:

A. Escherichia coli bacteria
B. Tobacco mosaic virus
C. Drosophila melanogaster
D. Amphibian oocytes

___ **38.** What is the primary function of transfer RNA (tRNA) in protein synthesis?

A. Carrying genomic instructions from nucleus to ribosomes
B. Delivering specific amino acids to the ribosome based on codon recognition
C. Forming the catalytic peptide bond between amino acids
D. Phosphorylating ribosomal subunits

___ **39.** Fossilized remains of Australopithecus afarensis ('Lucy' or 'Dinkinesh') discovered in the Afar depression of Ethiopia provided crucial evidence for:

A. The evolution of modern industrial tools
B. Early obligate bipedal locomotion in hominid evolution
C. The invention of agricultural cultivation
D. Neanderthal gene flow into African populations

___ **40.** Which of the following statements correctly characterizes an enzyme-catalyzed reaction?

A. Enzymes increase the activation energy of the transition state
B. Enzymes are consumed and depleted during the biochemical reaction
C. Enzymes decrease activation energy without altering the overall free energy change (ΔG)
D. Enzymes shift the chemical equilibrium constant toward irreversible products